

FeCode_updated2025_20260311

2026-03-12

Things that need to be changed

```
Date_Run = "20260311"
plates<- c("Plate15", "Plate16", "Plate17")
Std_plates<- c("STD")
Run_by = "Melanie Giessner and Zoe Read"#Instrument user
Script_run_by = "Zoe Read"#Code user
Project = "COMPASS_Synoptic_CB"
Experiment="2025"
Run_notes=""
The Fe2 diff std curve R squared is 0.94, which is below the 0.985 cutoff but I think is reasonable,
and Melanie mentioned that this R squared might be lower since it is the difference between two curves.
The FE_2_high std curve R squared is 0.984, which is close to the 0.985 cutoff
so I think it is reasonable.
The FE_2_diff slope is slightly lower than previous slopes.
The check stds for FE3 60 uM and Fe3 9 uM were all bad.
One dup was off for Fe2 and one dup was off for Fe3.
One spike was too high and one spike looks like the spk solution didn't get added. "#any notes from run

#Stds that should be excluded
stds_to_remove<-c("45 uM Fe_2 B")
```

File paths

```
## [1] "Tidy Data/20260311_COMPASS_SynopticCB_Fe_STD.csv"

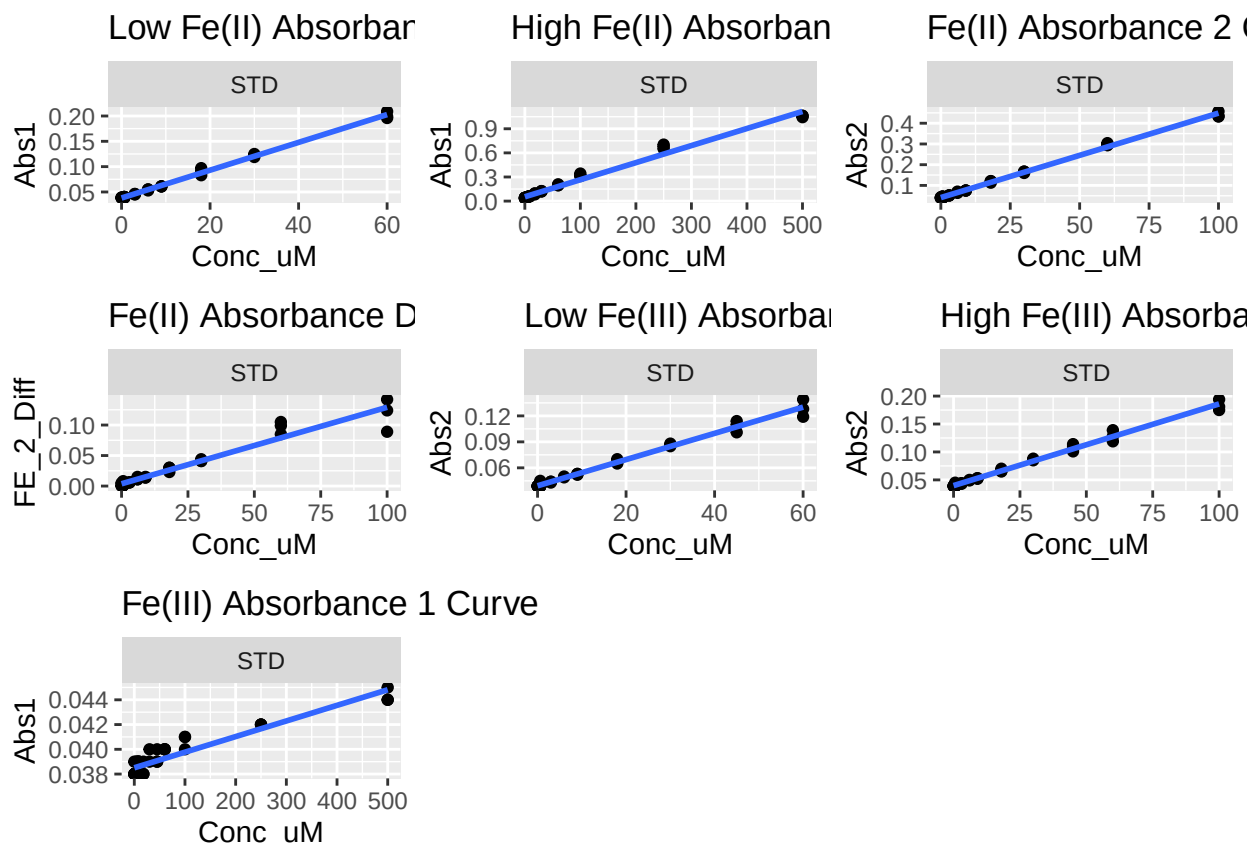
## [1] "Tidy Data/20260311_COMPASS_SynopticCB_Fe_Plate15.csv"
## [2] "Tidy Data/20260311_COMPASS_SynopticCB_Fe_Plate16.csv"
## [3] "Tidy Data/20260311_COMPASS_SynopticCB_Fe_Plate17.csv"
```

Plot standards

Table 1: Slopes

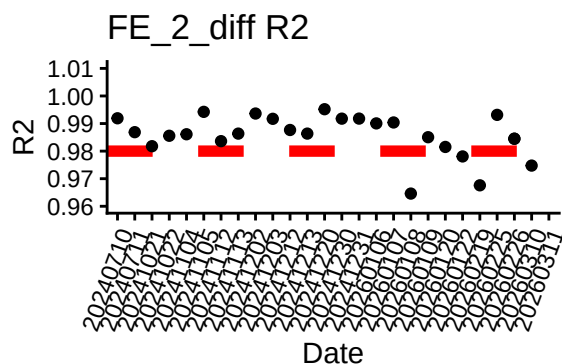
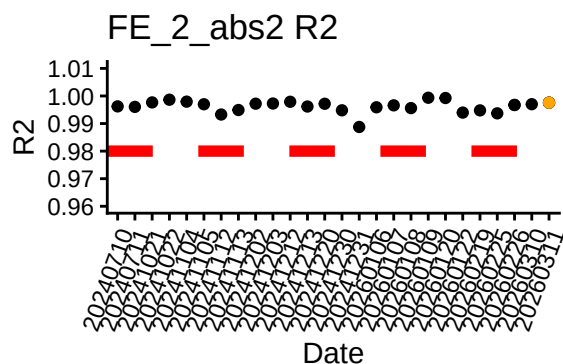
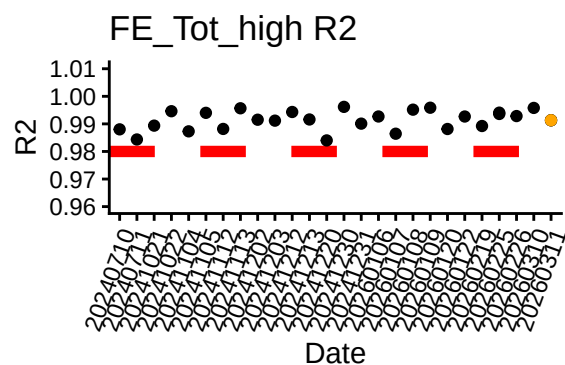
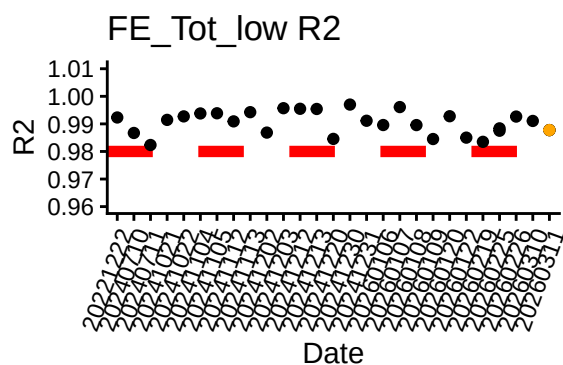
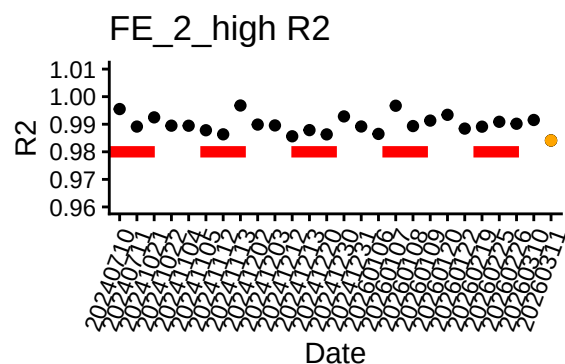
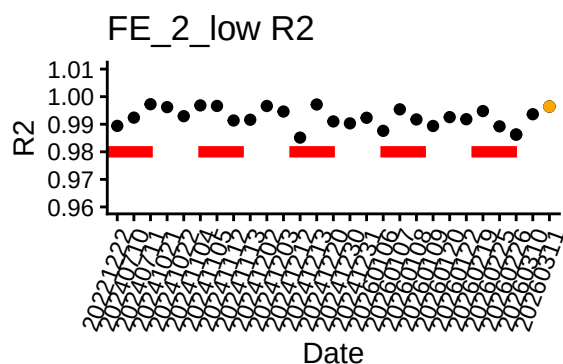
Date	Project	Curve	R2	Slope	Intercept	Top_STD
20260311	COMPASS_Synoptic_CB	FE_2_low	0.9964268	0.0027444	0.0381174	60
20260311	COMPASS_Synoptic_CB	FE_2_high	0.9841118	0.0021245	0.0535303	500
20260311	COMPASS_Synoptic_CB	FE_Tot_low	0.9877554	0.0015136	0.0392979	60

Date	Project	Curve	R2	Slope	Intercept	Top_STD
20260311	COMPASS_Synoptic_CFE_Tot_high		0.9913145	0.0014617	0.0398036	100
20260311	COMPASS_Synoptic_CFE_2_abs2		0.9975544	0.0040783	0.0411820	100
20260311	COMPASS_Synoptic_CFE_3_abs1		0.8873555	0.0000126	0.0385052	500
20260311	COMPASS_Synoptic_CFE_2_diff		0.9400702	0.0012542	0.0037123	100



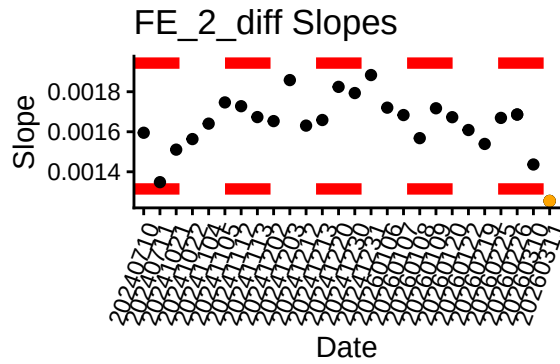
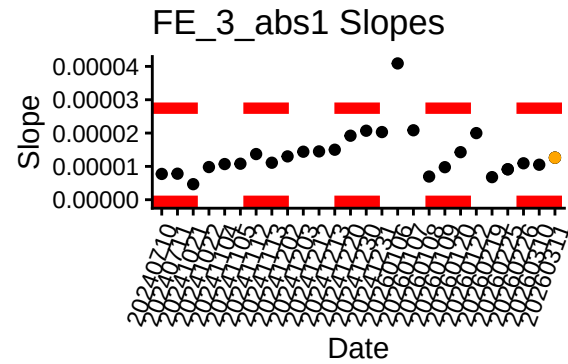
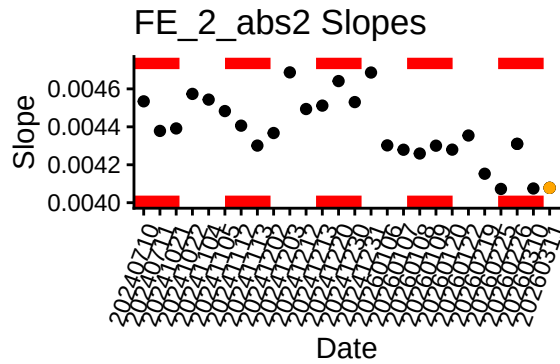
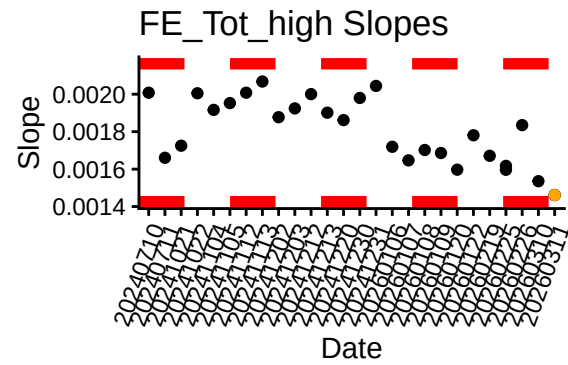
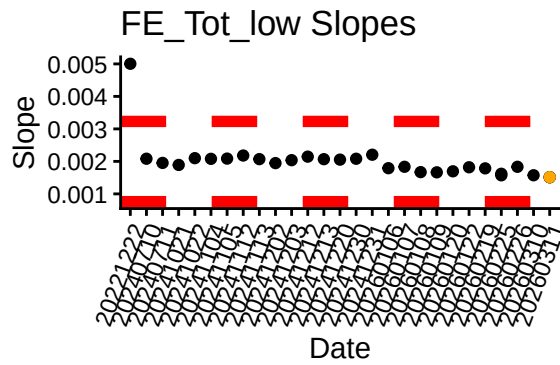
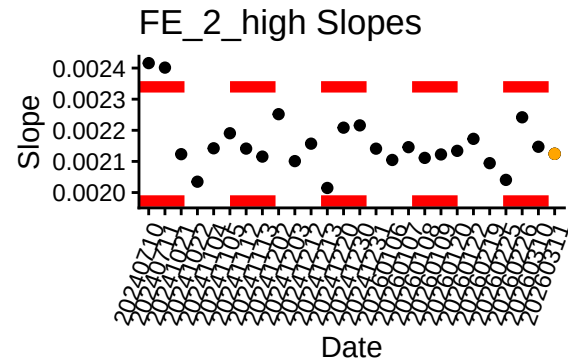
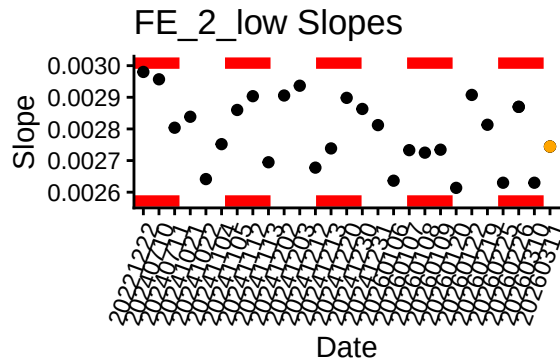
Checking STD Data against QAQC file

R2



```
## [1] "FE_2_low: Std Curve r2 GOOD"
## [1] "FE_2_high: Std Curve r2 is below cutoff! - REASSESS"
## [1] "FE_Tot_low: Std Curve r2 GOOD"
## [1] "FE_Tot_high: Std Curve r2 GOOD"
## [1] "FE_2_abs2: Std Curve r2 GOOD"
## [1] "FE_3_abs1: R2 will always be low"
## [1] "FE_2_diff: Std Curve r2 is below cutoff! - REASSESS"
```

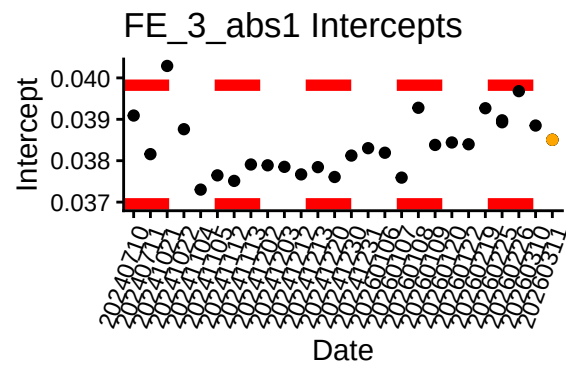
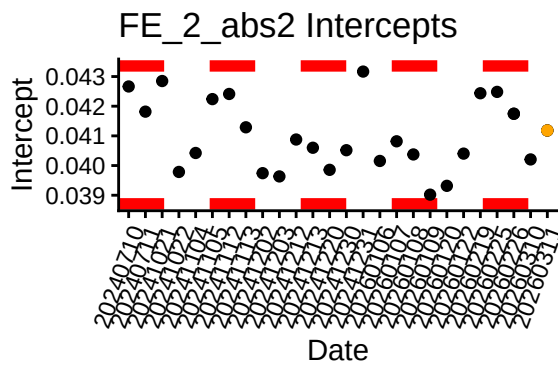
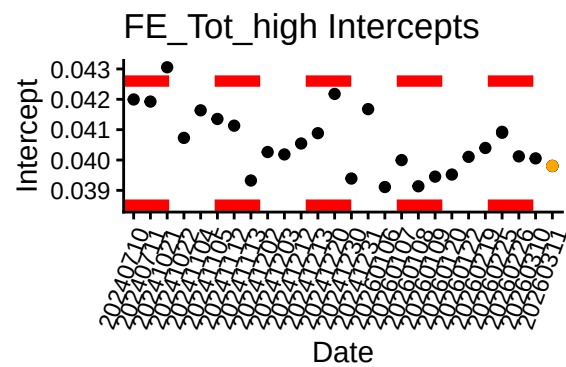
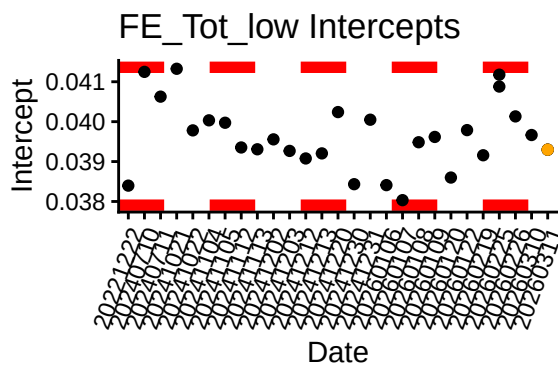
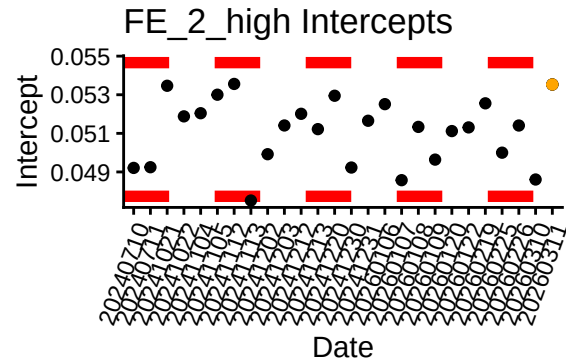
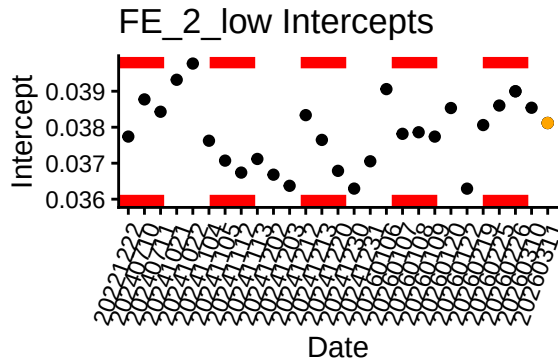
Slopes

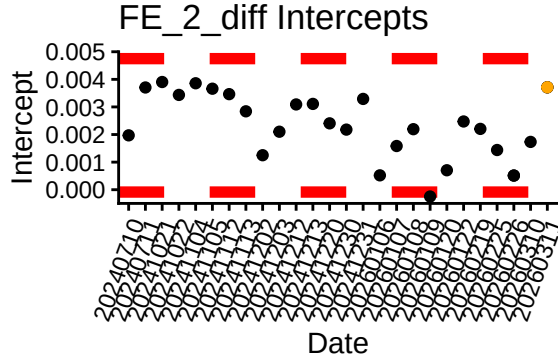


```
## [1] "FE_2_low:Std curve slope is with 2 sd of previous slopes"
## [1] "FE_2_high:Std curve slope is with 2 sd of previous slopes"
```

```
## [1] "FE_Tot_low:Std curve slope is with 2 sd of previous slopes"
## [1] "FE_Tot_high:Std curve slope is with 2 sd of previous slopes"
## [1] "FE_2_abs2:Std curve slope is with 2 sd of previous slopes"
## [1] "FE_3_abs1:Std curve slope is with 2 sd of previous slopes"
## [1] "FE_2_diff: Std curve slope is 2 sd different from previous slopes! \n      - REASSESS"
```

Intercepts



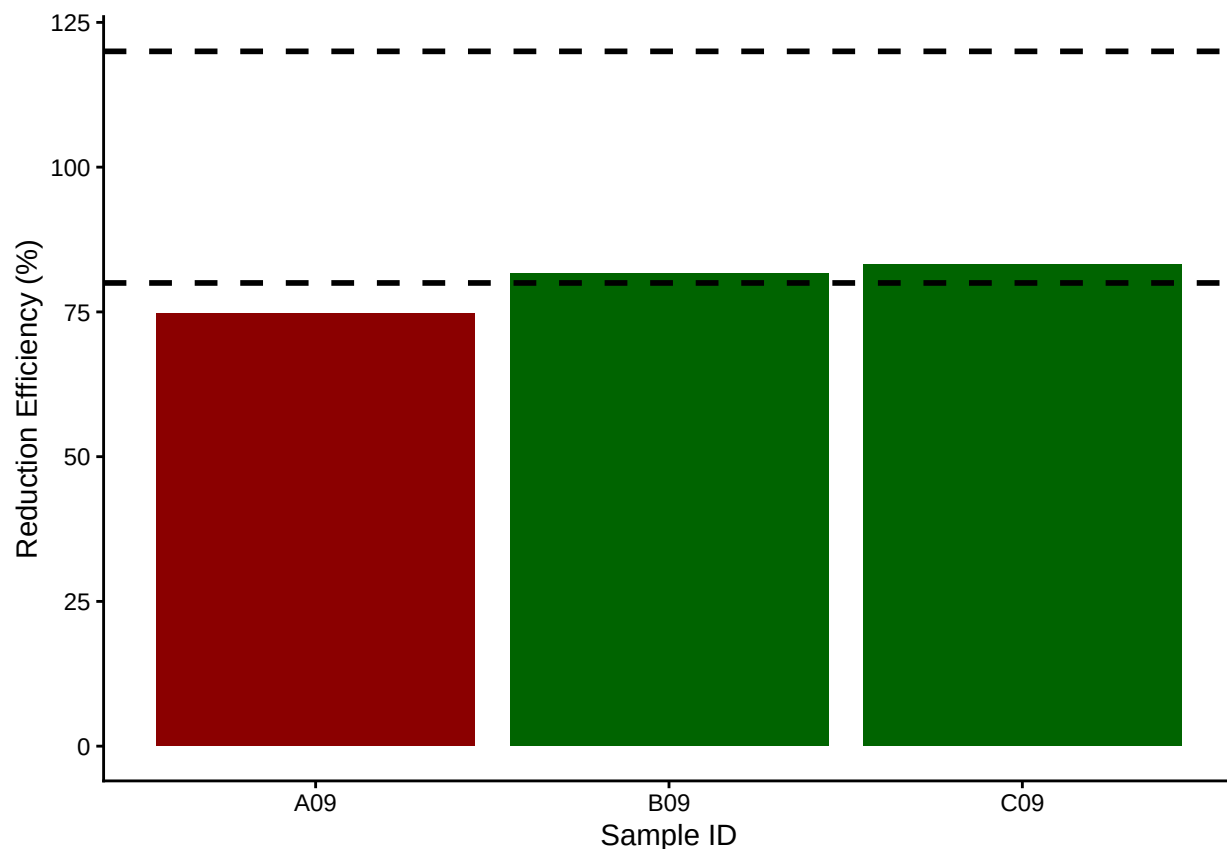


```
## [1] "FE_2_low:Std curve intercept is with 2 sd of previous intercepts"
## [1] "FE_2_high:Std curve intercept is with 2 sd of previous intercepts"
## [1] "FE_Tot_low:Std curve intercept is with 2 sd of previous intercepts"
## [1] "FE_Tot_high:Std curve intercept is with 2 sd of previous intercepts"
## [1] "FE_2_abs2:Std curve intercept is with 2 sd of previous intercepts"
## [1] "FE_3_abs1:Std curve intercept is with 2 sd of previous intercepts"
## [1] "FE_2_diff:Std curve intercept is with 2 sd of previous intercepts"
```

Calculate Reduction Efficiency

Table 2: Reduction Efficiency

ID	FE2	FE3	Eff	Eff1	Eff_flag
A09	0.199	0.119	59.79899	74.74874	NO, rerun
B09	0.196	0.128	65.30612	81.63265	OK
C09	0.209	0.139	66.50718	83.13397	OK



```
## [1] ">60% of Reduction Efficiencies are within range"
```

Method Minimum Detection Limit (MDL) Calculation

I use the lowest standard to calculate MDL How to calculate MDL https://www.epa.gov/sites/default/files/2016-12/documents/mdl-procedure_rev2_12-13-2016.pdf Look at Table 1 on page 5 The Student's t-value used to calculate the method detection limit (MDL) is the one appropriate for a 99% confidence level and a standard deviation estimate with n-1 degrees of freedom

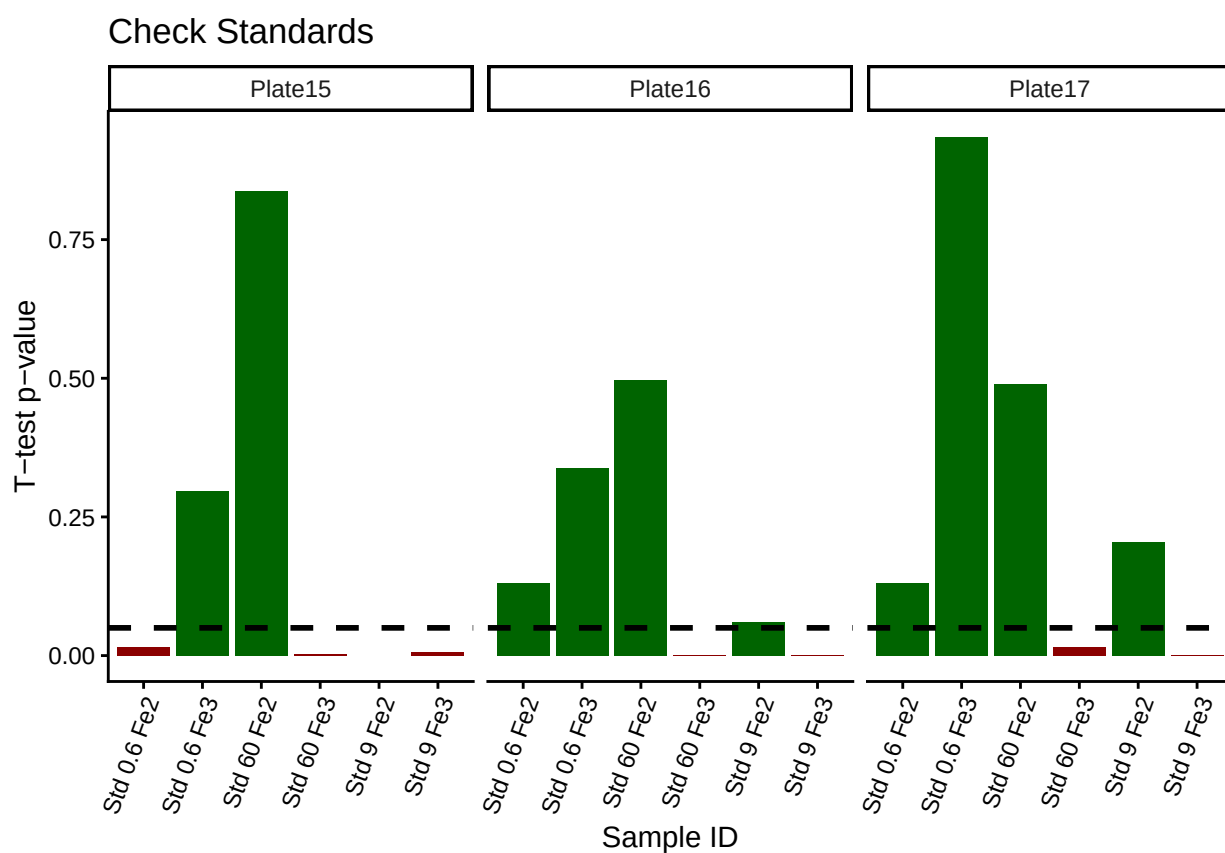
```
## [1] 0.001419471
```

Check standards QAQC

```
## [1] "Plate15"
## [1] "Fe2 Chk Std 0.6 is significantly different from Std - REASSESS"
## [1] "Std excluded"
## [1] "Fe2 Chk Std 60 GOOD"
## [1] "Fe 3 Chk Std 0.6 Fe2 GOOD"
## [1] "Fe 3 Chk Std 9 is significantly different from Std - REASSESS"
## [1] "Fe 3 Chk Std 60 is significantly different from Std - REASSESS"
## [1] "Plate16"
## [1] "Fe2 Chk Std 0.6 GOOD"
## [1] "Fe2 Chk Std 9 GOOD"
```

```
## [1] "Fe2 Chk Std 60 GOOD"
## [1] "Fe 3 Chk Std 0.6 Fe2 GOOD"
## [1] "Fe 3 Chk Std 9 is significantly different from Std - REASSESS"
## [1] "Fe 3 Chk Std 60 is significantly different from Std - REASSESS"
## [1] "Plate17"
## [1] "Fe2 Chk Std 0.6 GOOD"
## [1] "Fe2 Chk Std 9 GOOD"
## [1] "Fe2 Chk Std 60 GOOD"
## [1] "Fe 3 Chk Std 0.6 Fe2 GOOD"
## [1] "Fe 3 Chk Std 9 is significantly different from Std - REASSESS"
## [1] "Fe 3 Chk Std 60 is significantly different from Std - REASSESS"
```

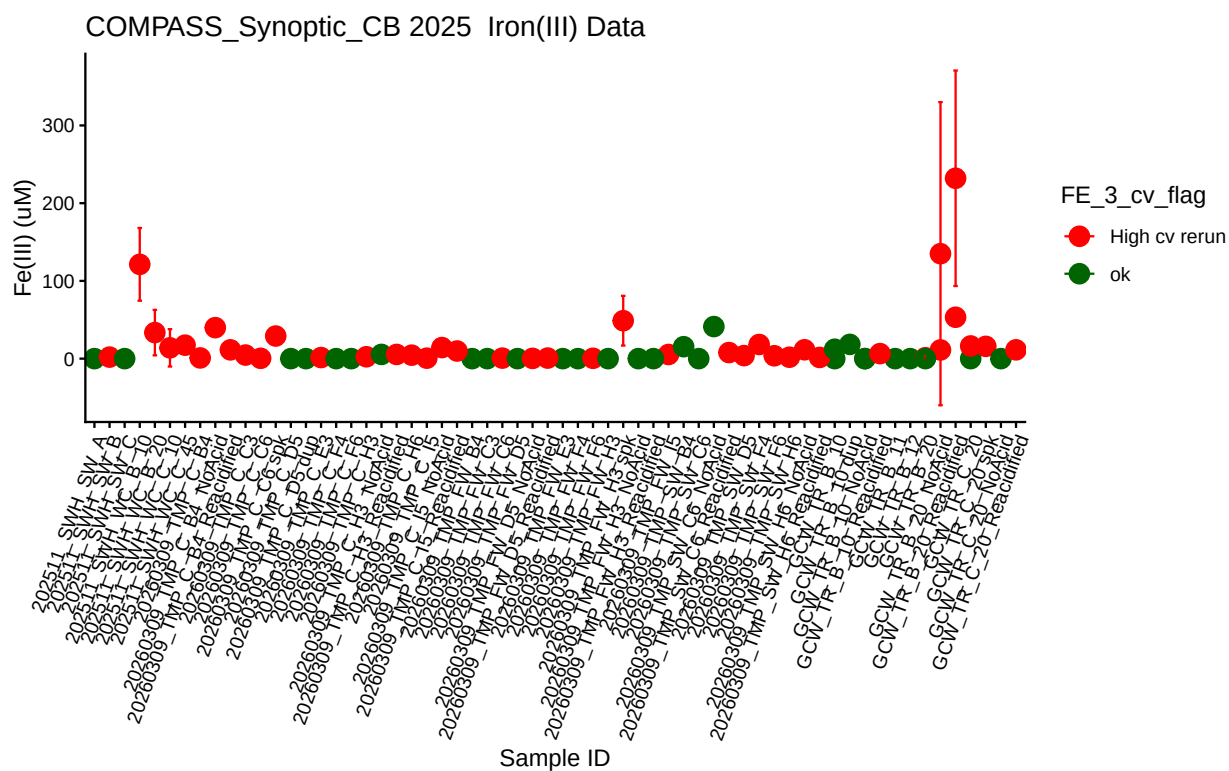
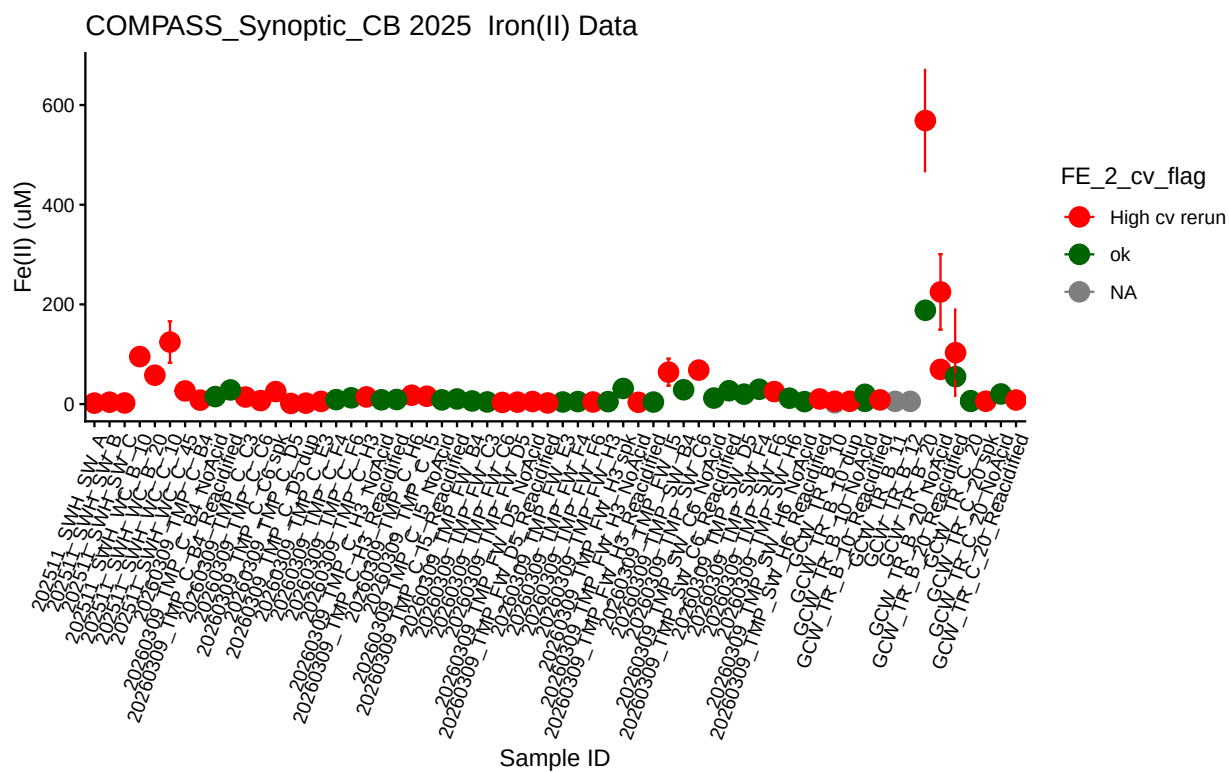
```
## Warning: Removed 1 row containing missing values or values outside the scale range
## ('geom_bar()').
```



#No Matrix Checks We do not use matrix checks for the Fe samples because our matrix checks are not made with Trace metal grade reagents. If we want to do this in the future, we need to use trace metal grade sodium chloride and sodium bicarbonate for making the matrix checks

Subset Data and Calculate Concentrations

Calculate Averages across wells, std. dev, and cv.



Remove Bad Reps

Flagged data

Table 3: High CV Samples

ID full	FE 2 mean	FE 2 cv	FE 3 mean	FE 3 cv	FE 2 cv flag	FE 3 cv flag
Plate15_1_202511_SWH_WC_B_10	100.952377	2.6376011	146.6322207	15.677018	ok	High cv rerun
Plate15_1_202511_SWH_WC_B_20	48.238084	5.8755244	50.2308194	10.712146	ok	High cv rerun
Plate15_1_202511_SWH_WC_C_10	102.129145	21.1835788	0.0000000	0.000000	High cv rerun	ok
Plate15_1_20260309_TMP_C_B4	7.062721	10.9444314	0.0000000	0.000000	High cv rerun	ok
Plate15_1_20260309_TMP_C_C6	7.973680	19.3881525	0.0000000	0.000000	High cv rerun	ok
Plate15_1_20260309_TMP_C_E3	6.516145	23.7249052	0.0000000	0.000000	High cv rerun	ok
Plate15_1_20260309_TMP_FW_F6	3.418884	7.5363263	0.6661319	32.319984	ok	High cv rerun
Plate16_1_20260309_TMP_C_H3	17.083274	12.0659877	0.0000000	0.000000	High cv rerun	ok
Plate16_1_20260309_TMP_C_I5	13.075053	15.7648751	0.0000000	0.000000	High cv rerun	ok
Plate16_1_20260309_TMP_FW_D5_NoAcid	1.596965	16.1342455	0.0000000	0.000000	High cv rerun	ok
Plate16_1_20260309_TMP_FW_H3 spk	31.294240	7.6353476	63.2389550	44.534219	ok	High cv rerun
Plate16_1_20260309_TMP_SW_D5	19.998344	4.8207430	5.8523738	28.869757	ok	High cv rerun
Plate16_1_20260309_TMP_SW_F4	29.715243	8.1647876	19.8617156	13.473164	ok	High cv rerun
Plate16_1_20260309_TMP_SW_F6	27.468210	6.5661631	5.5164928	84.130201	ok	High cv rerun
Plate16_1_20260309_TMP_SW_H6	11.860440	9.8759391	2.4766547	52.157545	ok	High cv rerun
Plate16_8_GCW_TR_B_20_NoAcid	63.789442	6.4627175	16.7618183	26.958871	ok	High cv rerun
Plate17_1_20260309_TMP_C_H3_Reacidified	9.188293	6.0577670	6.7875600	11.306397	ok	High cv rerun
Plate17_1_20260309_TMP_C_I5_Reacidified	10.281444	5.4136886	12.2034312	28.455115	ok	High cv rerun
Plate17_1_20260309_TMP_FW_D5_Reacidified	2.507924	0.0000000	1.0655409	128.835399	ok	High cv rerun
Plate17_1_20260309_TMP_SW_C6_Reacidified	26.800173	4.1537444	9.3323414	17.035348	ok	High cv rerun
Plate17_1_GCW_TR_B_10_Reacidified	7.062721	10.9444314	8.6143878	56.716857	High cv rerun	High cv rerun
Plate17_1_GCW_TR_B_20_NoAcid	268.759427	0.3715294	202.5276893	108.975559	ok	High cv rerun
Plate17_1_GCW_TR_B_20_Reacidified	53.157264	0.9694186	310.9673106	10.500066	ok	High cv rerun
Plate17_1_GCW_TR_C_20_Reacidified	6.698337	19.2329984	8.9671662	24.009114	High cv rerun	High cv rerun
Plate17_1_GCW_TR_B_10 dup	5.058610	10.1869169	18.3316146	3.817414	High cv rerun	ok

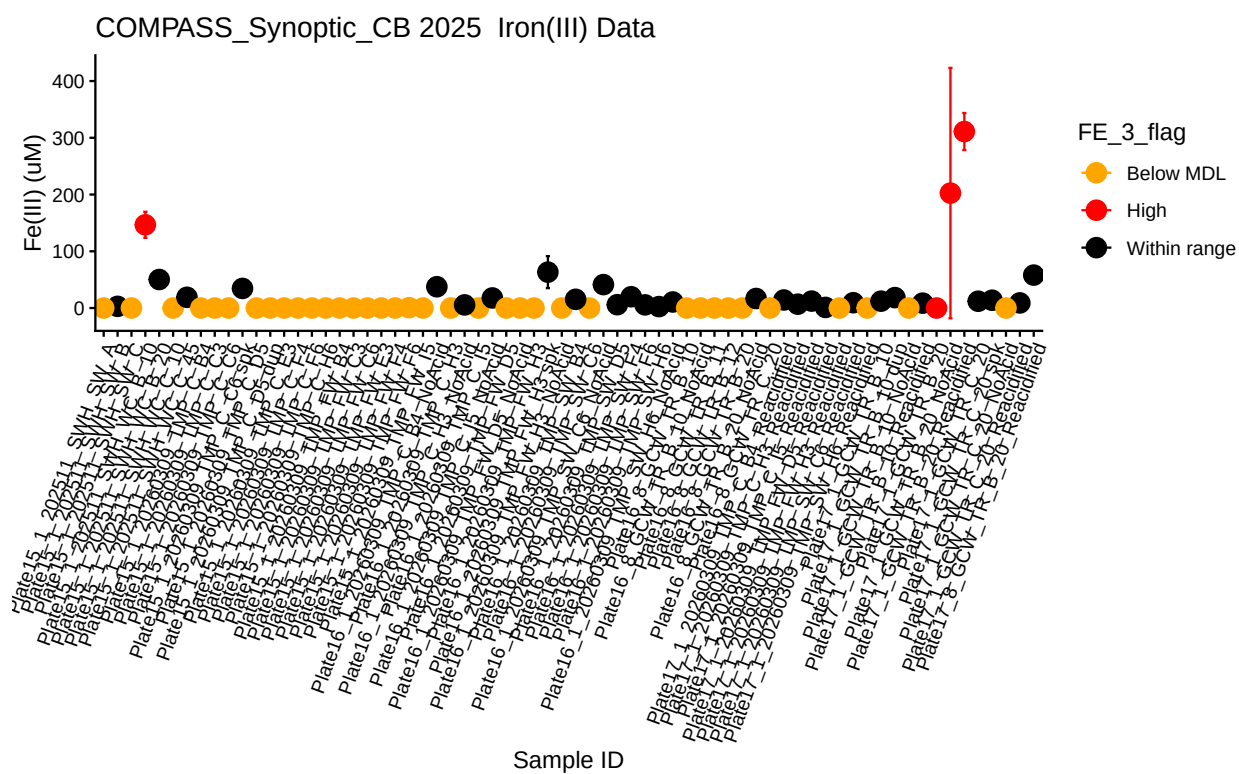
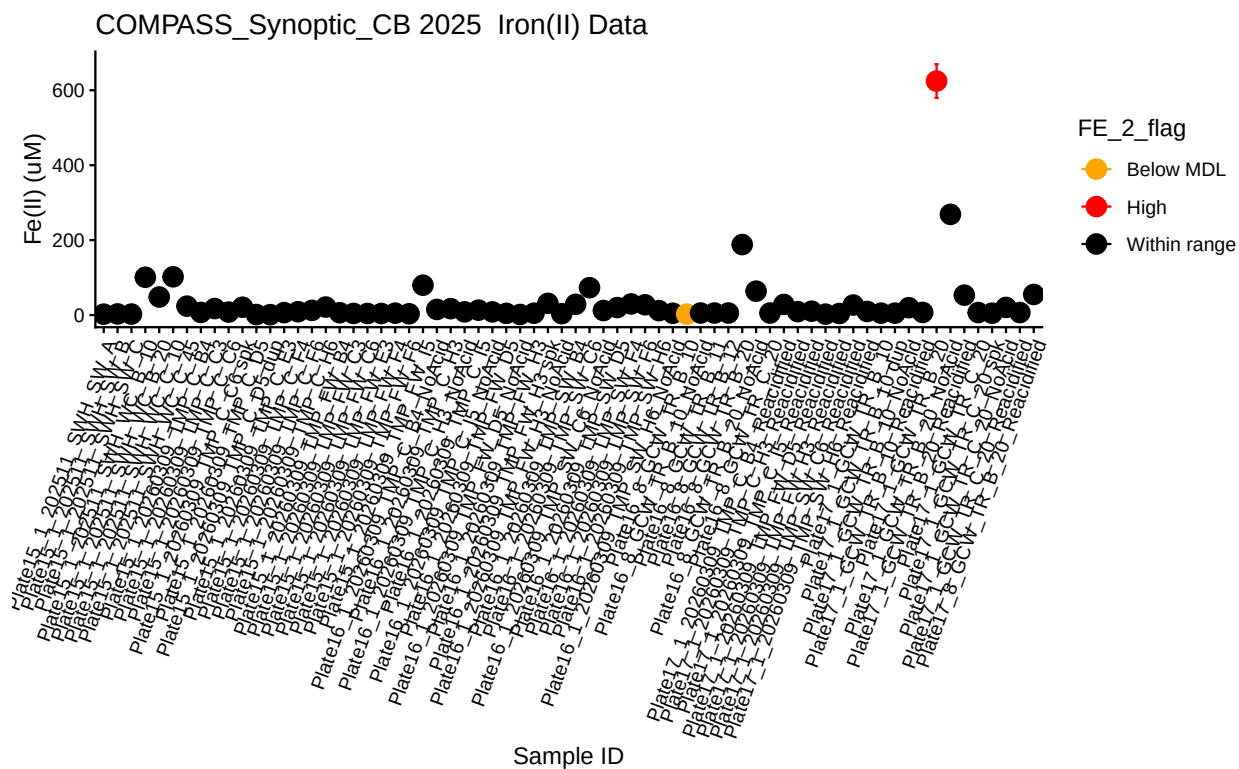
Table 4: Samples Above the Detection Limit that should be rerun at a higher dilution

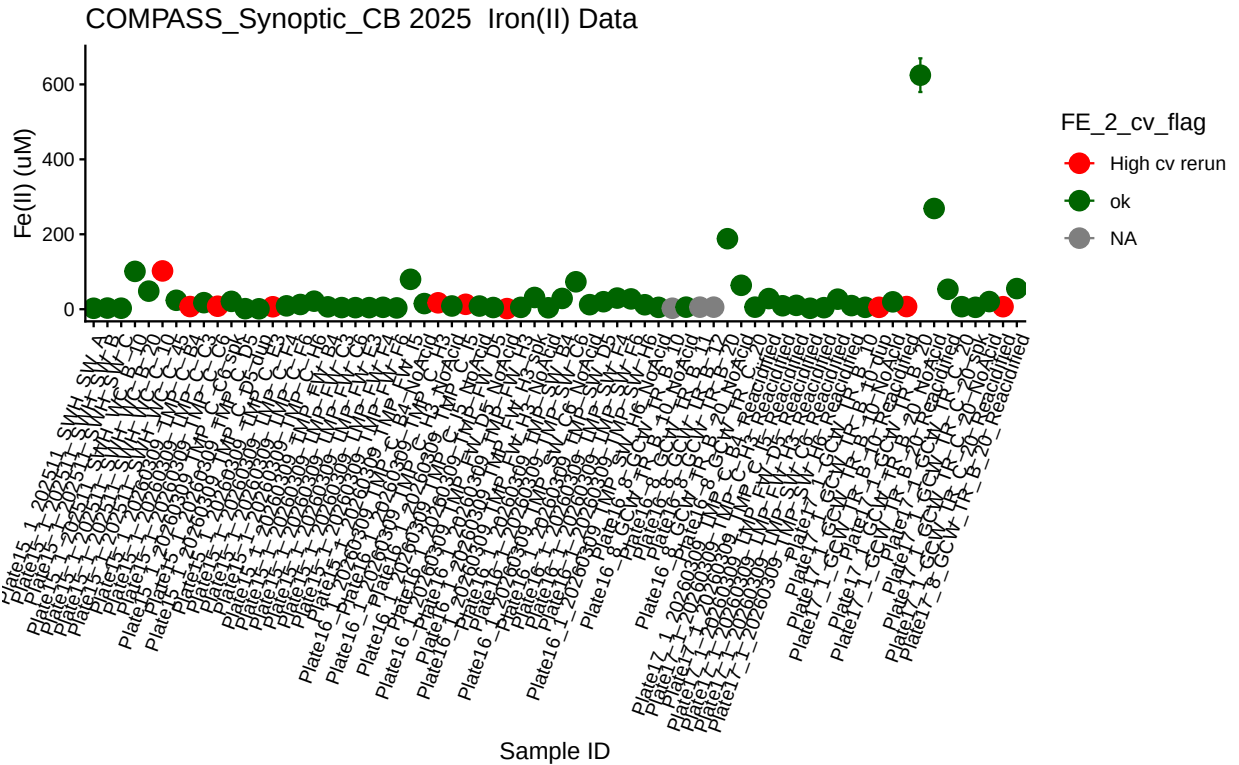
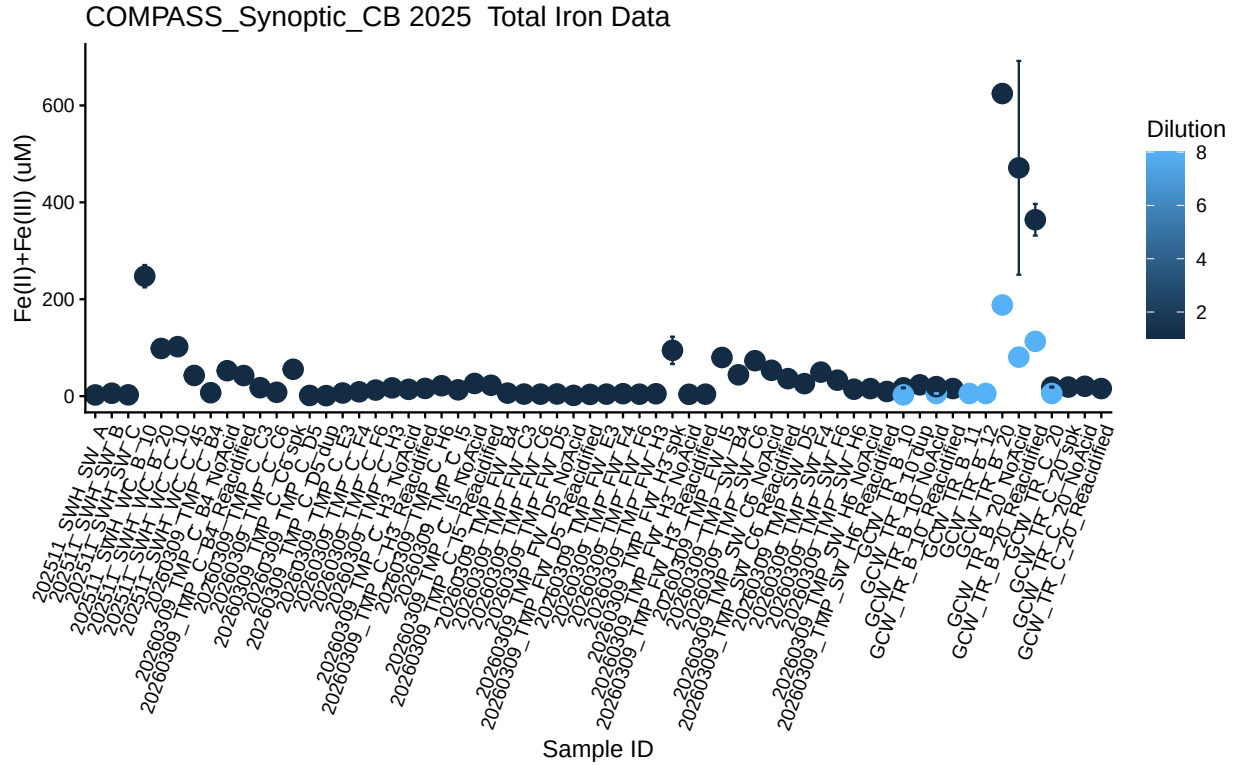
ID full	FE 2 mean	FE 3 mean	FE tot mean	FE 2 flag	FE 3 flag
Plate15_1_202511_SWH_WC_B_10	100.9524	146.6322	247.5846	Within range	High

Table 5: Samples Below the Detection Limit that should be rerun at a lower dilution

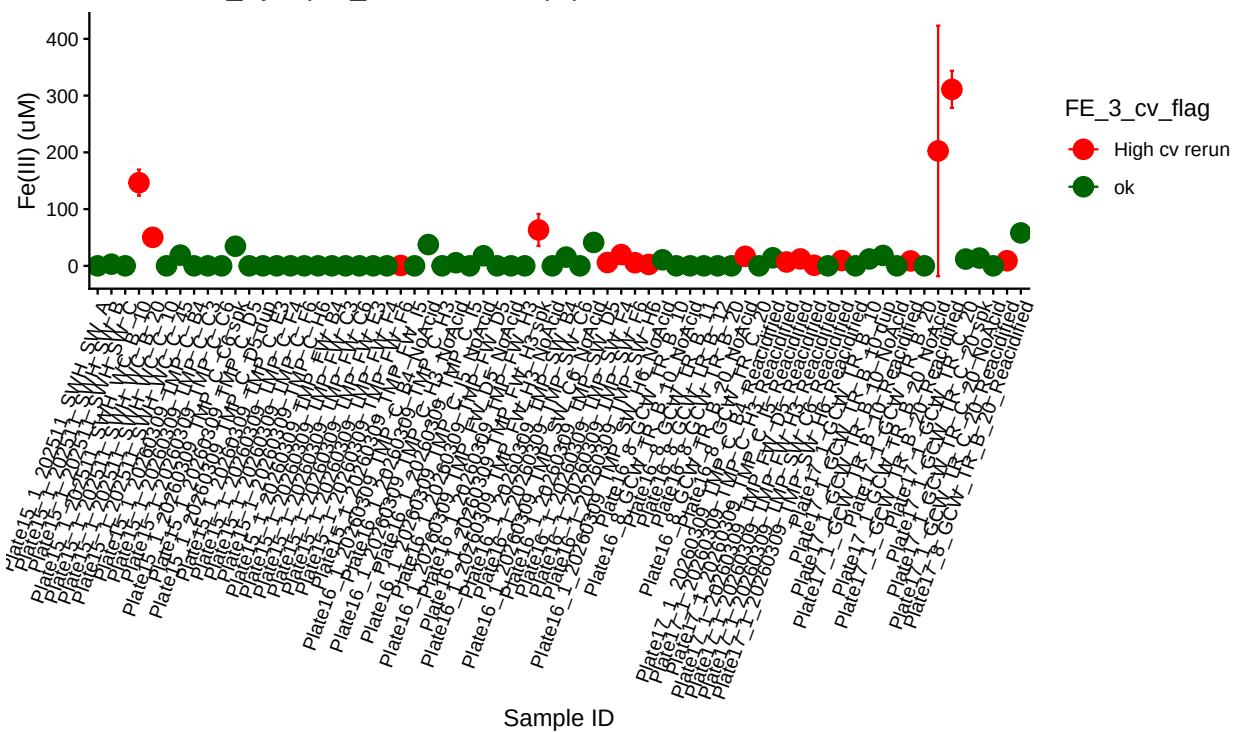
ID full	FE 2 mean	FE 3 mean	FE tot mean	FE 2 flag	FE 3 flag
Plate16_8_GCW_TR_B_10_NoAcid	5.488043	0	5.488043	Within range	Below MDL
Plate16_8_GCW_TR_B_11	5.488043	0	5.488043	Within range	Below MDL
Plate16_8_GCW_TR_B_12	5.488043	0	5.488043	Within range	Below MDL

Plot data after bad reps were removed

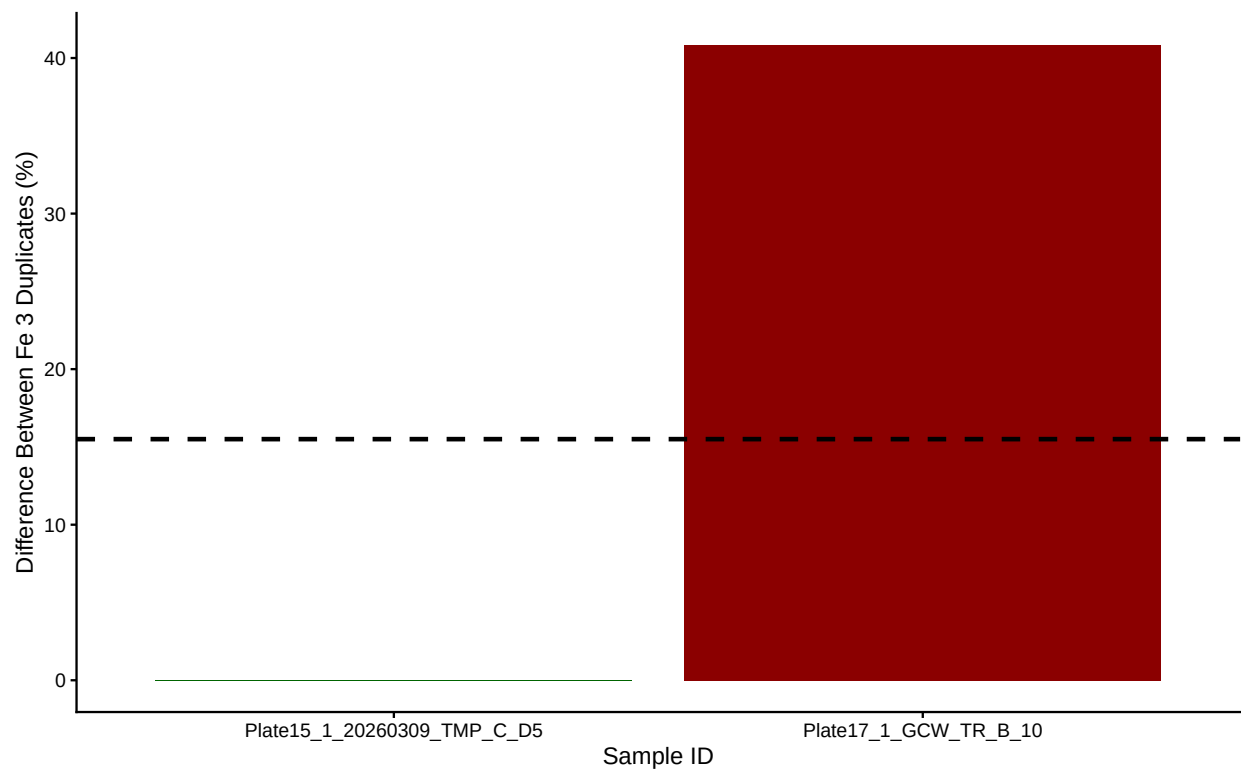
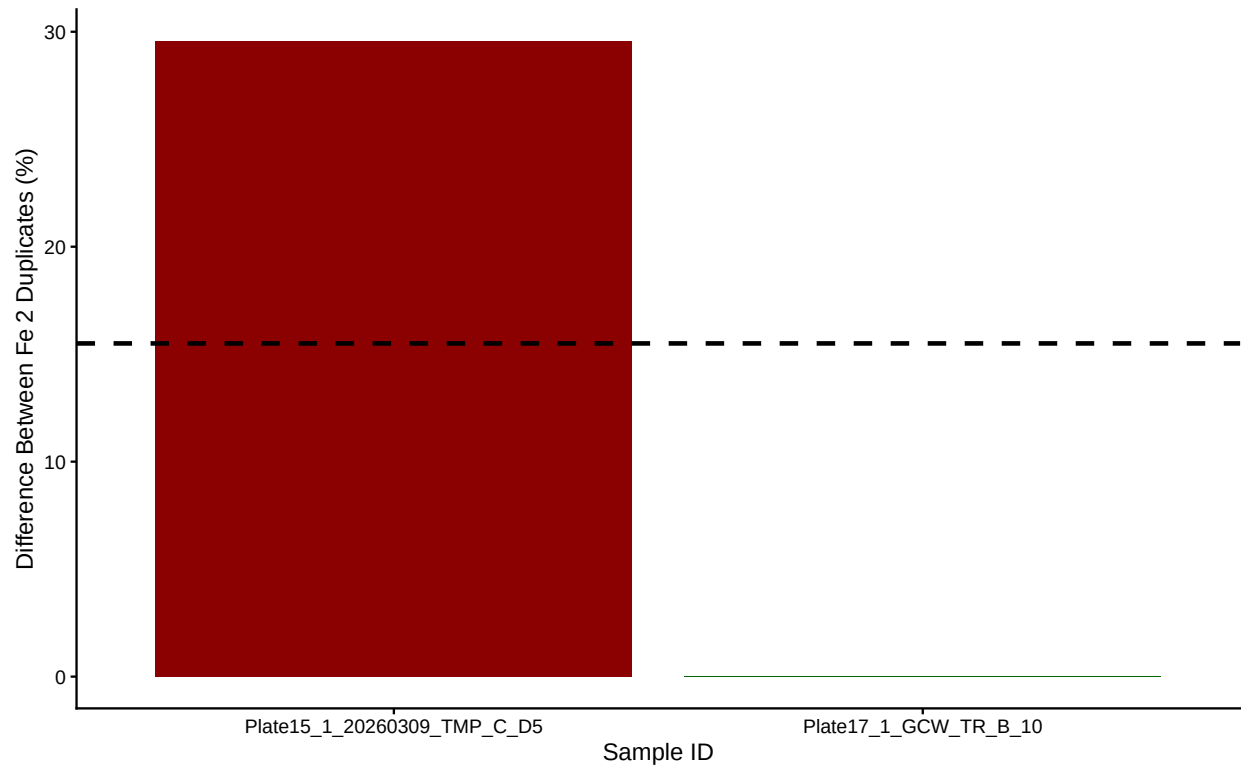


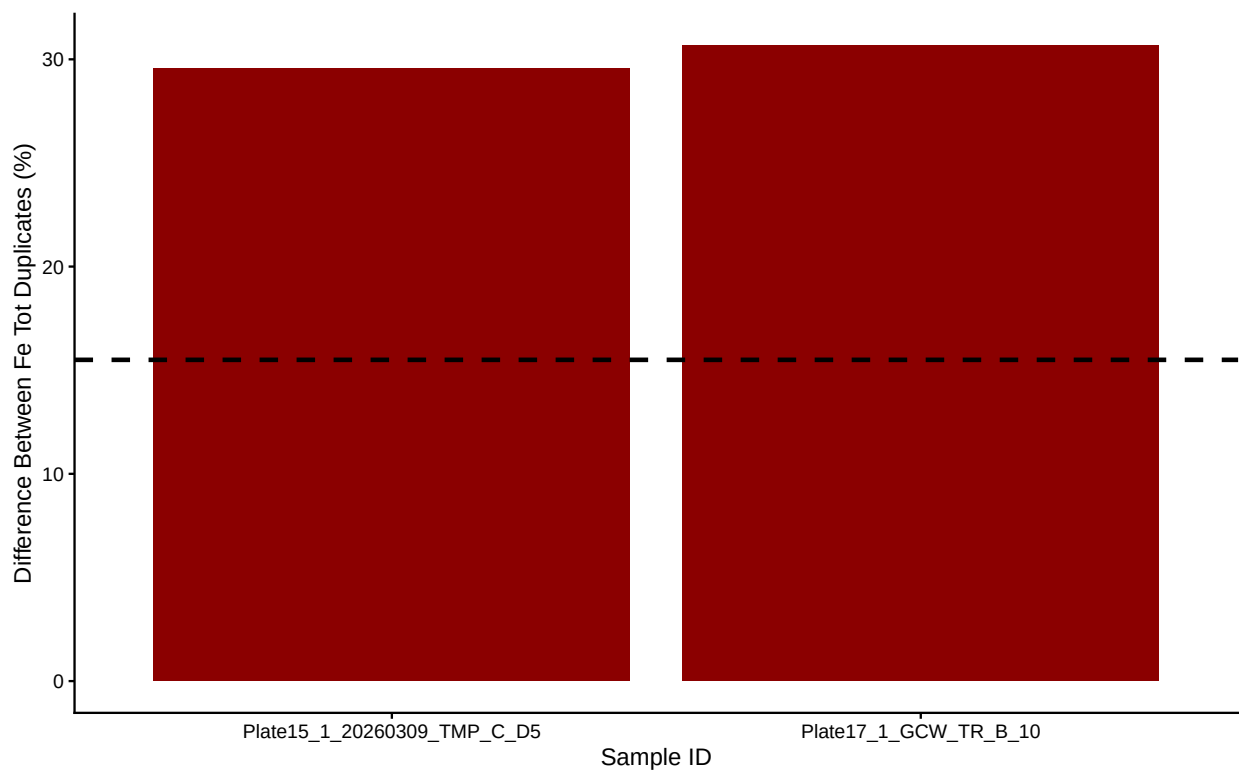


COMPASS_Synoptic_CB 2025 Iron(III) Data



Check Dups for QAQC

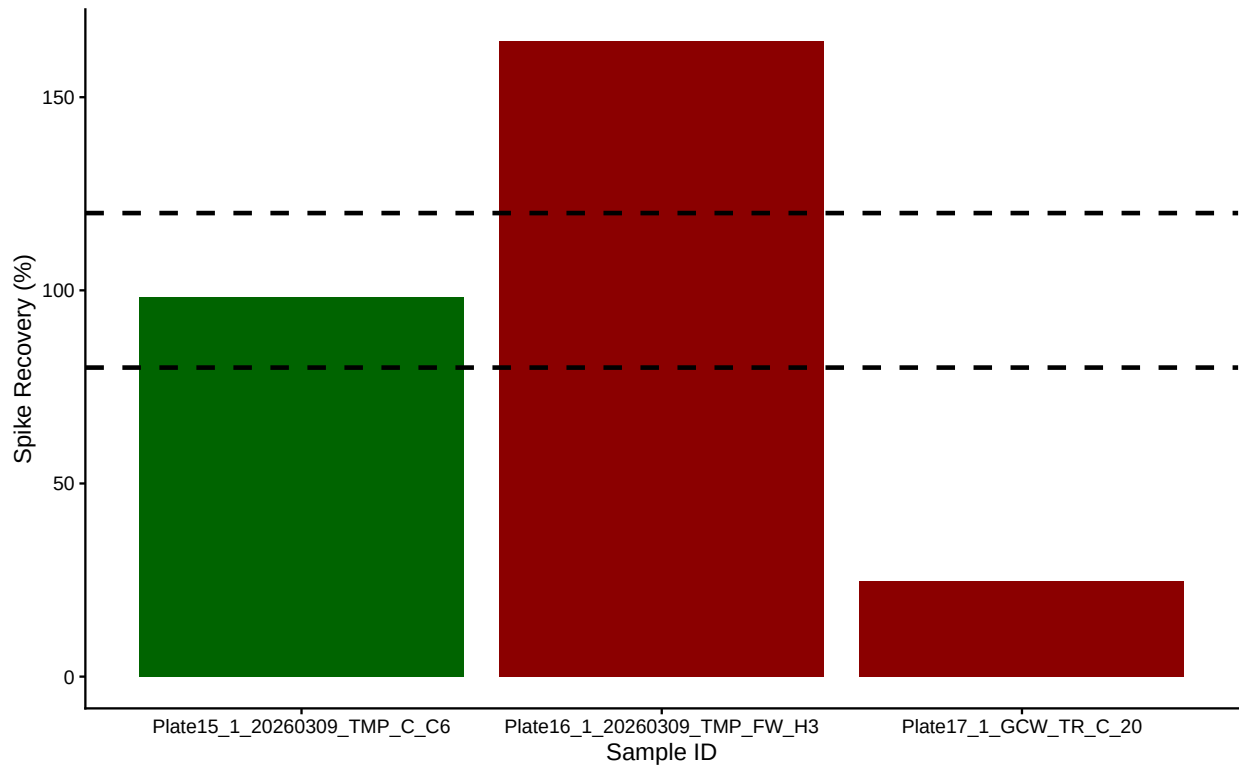




```
## [1] "<60% of Duplicates are within <10% - REASSESS"
```

```
## [1] "<60% of Duplicates are within <10% - REASSESS"
```

Check the spks for QAQC



```
## [1] "<60% of Spikes are out of range - REASSESS"
```

Export full data then just final data